

QCMS Enterprise OS

Quality Circle & Compliance Management System (IFQM)

Comprehensive Technical Knowledge Transfer (KT), System Architecture, Subsystem Specifications, Data Entity Relationships & Operational Runbook

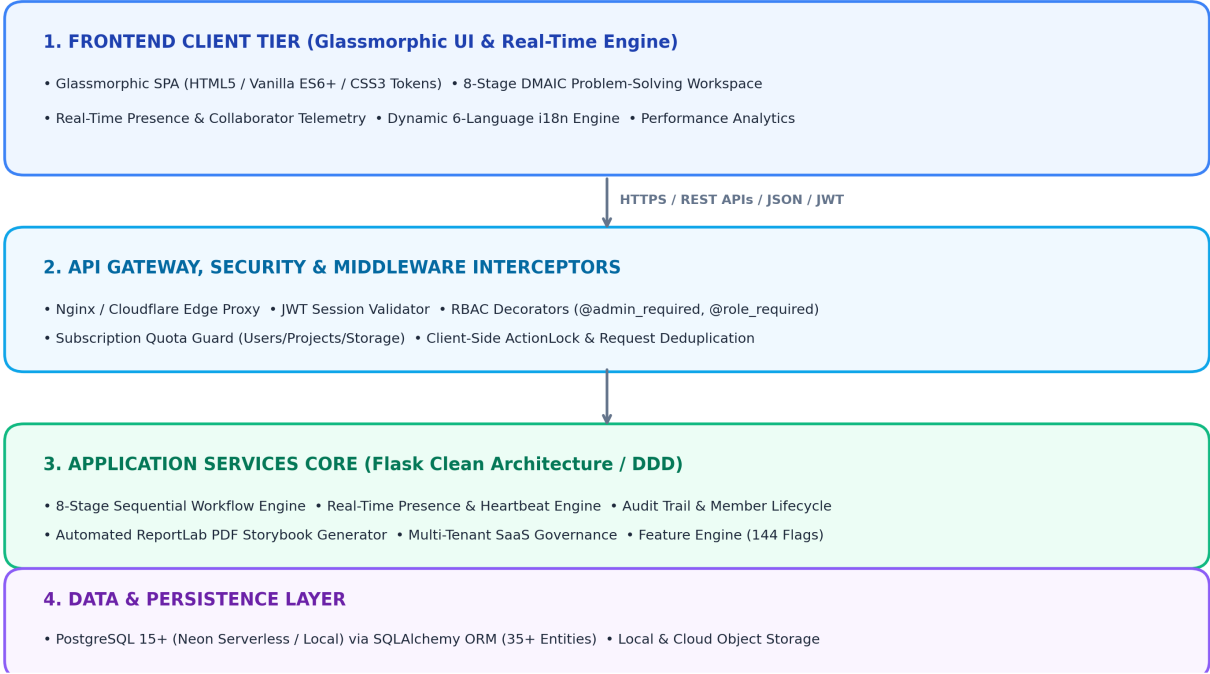
Document Version:	2.6.0 (Enterprise Release)
Target Audience:	Engineering Leads, Solutions Architects, Full-Stack Developers, QA & DevOps Teams
Platform Sponsor:	Indian Foundation for Quality Management (IFQM)
Core Frameworks:	Flask Clean Architecture (Python 3.10+), PostgreSQL 15, Vanilla ES6+ Glassmorphic Frontend
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Executive Summary:

QCMS Enterprise OS is an industrial-grade, multi-tenant quality governance platform designed to guide manufacturing circles through rigid 8-Stage DMAIC problem solving, real-time presence collaboration, forensic member transition audits, ReportLab-powered QC storybook publishing, and multi-organization tenant isolation with granular 144-module feature flags.

1. High-Level System Architecture & Component Topology

QCM follows Clean Architecture and Domain-Driven Design (DDD) principles. The architecture is cleanly partitioned into four isolated tiers: Client Presentation Tier, Gateway/Security Interceptor Tier, Application Core Services, and Data/Persistence Storage.

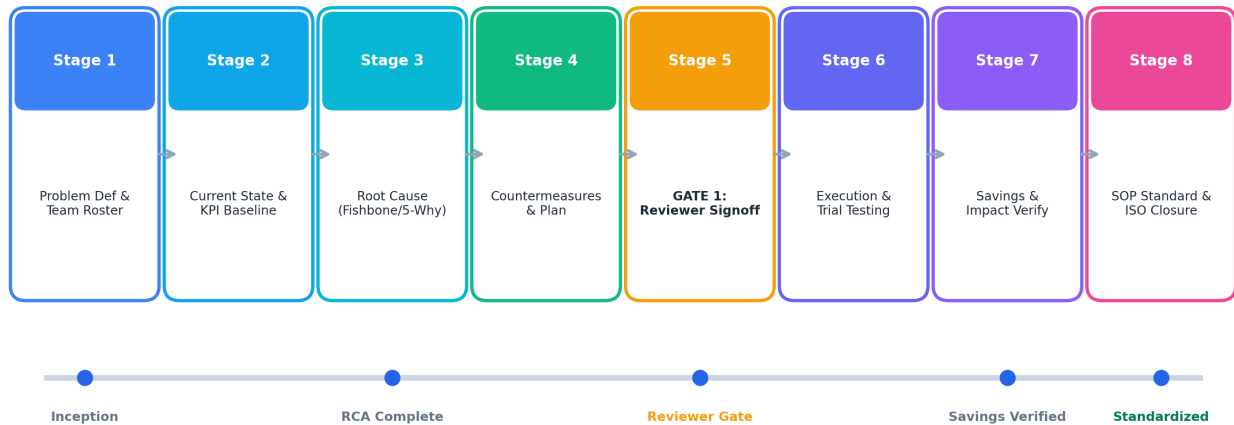


Tier-by-Tier Architectural Highlights:

- **Tier 1: Frontend Client (Glassmorphic SPA):** Built with lightweight, modern Vanilla ES6+ JavaScript, CSS3 custom properties (design system tokens), Bootstrap 5 grid layout, and Lucide icons. Includes real-time presence heartbeats, dynamic 6-language i18n DOM translation observer, ActionLock write deduplication, and responsive role-based dashboards.
- **Tier 2: API Gateway & Security Interceptor Layer:** Proxied through Nginx/Cloudflare for SSL termination, DDoS protection, and rate limiting. Enforces cryptographic JWT validation, Role-Based Access Control (RBAC), multi-tenant subscription quota verifiers (max users, projects, storage), and request idempotency.
- **Tier 3: Application Core Services (Flask Clean Architecture):** Hosts the 8-Stage sequential workflow state machine, live collaborator presence engine, forensic audit trail with tenure duration calculation, ReportLab PDF generators, and 144-module feature flag engine.
- **Tier 4: Data & Persistence Layer:** PostgreSQL database (Neon Serverless / Local) holding 35+ domain models with foreign keys, JSONB state representations, and encrypted password vaults, backed by secure filesystem and cloud asset storage.

2. 8-Stage DMAIC Quality Circle Workflow Engine

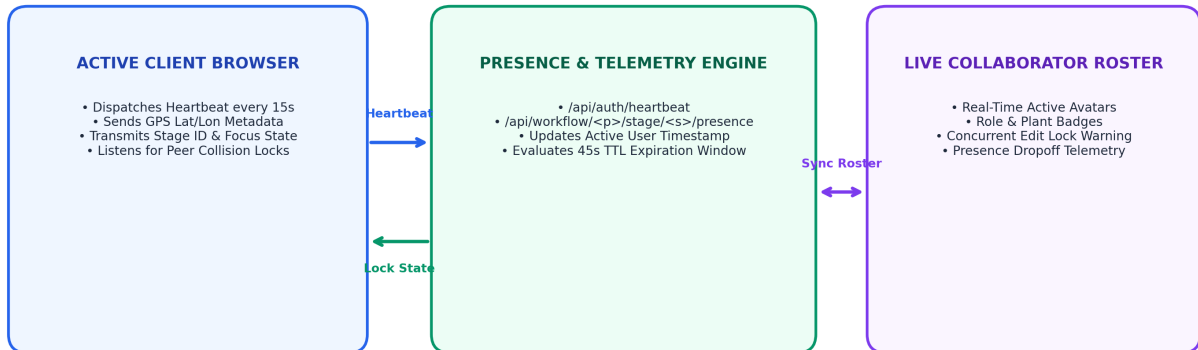
The heart of QCMS is a sequential problem-solving engine adhering to Quality Circle, 8D, DMAIC, and Six Sigma industrial methodologies. Each project progresses through 8 strictly locked stages, with independent gatekeeper approvals ensuring total compliance.



Stage	Title & Methodological Objective	Key Artifacts & Technical Logic	Approval Gate
Stage 1	Problem Definition & Circle Formation	5W2H framing, baseline metrics, Circle Leader, Facilitator, Reviewer, and Member roster assignment.	Leader Signoff
Stage 2	Current State & Baseline Data	KPI baseline measurements, data stratification, SOP deviation analysis, and physical photo evidence.	Leader Sync
Stage 3	Root Cause Analysis (RCA)	Interactive 6M Ishikawa (Fishbone) diagram, 5-Why analysis trees, and Pareto 80/20 frequency charts.	Facilitator Note
Stage 4	Countermeasure Planning	Action plans, cost-benefit analysis, barrier risk assessment, and financial ROI estimation.	Leader Submit
Stage 5	Independent Review & Approval	Gatekeeper signoff by designated Reviewer/Facilitator. Supports Approved, Revision, and Rejection states.	Reviewer Gate
Stage 6	Implementation & Trial Runs	Gantt milestone schedules, task matrices, pilot trial evaluation, and post-countermeasure telemetry.	Leader Update
Stage 7	Impact & Savings Verification	Before vs. After KPI improvement % calculations, tangible and intangible monetary savings verification.	Admin / QA Verify
Stage 8	Standardization & SOP Integration	Institutional SOP updates, lessons learned repository, final circle closure, and ISO 9001 certificate generation.	Final Closure

3. Real-Time Presence, Heartbeat & Concurrency Architecture

QCM Superpresence Collaboration across plant floors, QCMSS incorporates an active presence telemetry subsystem that prevents conflicting concurrent stage edits while providing live collaborator visibility.

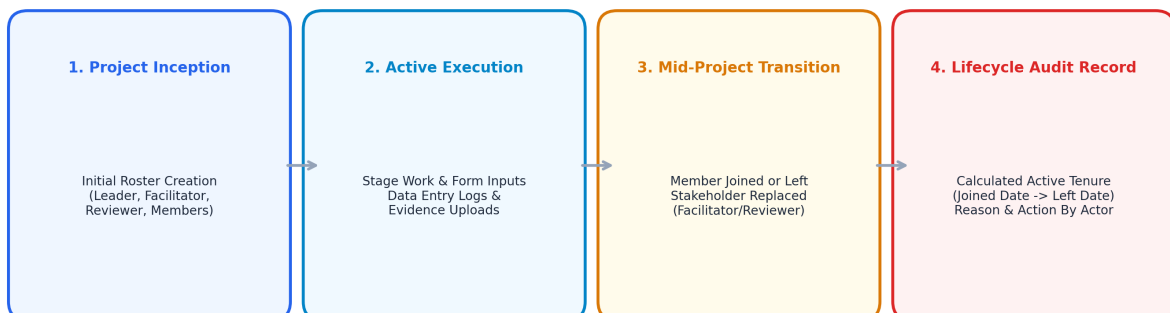


Presence Engine Mechanics:

- **Dual-Heartbeat Loop:** The frontend client dispatches global session heartbeats (`/api/auth/heartbeat`) every 30 seconds and stage-specific presence heartbeats (`/api/workflow//stage//presence-heartbeat`) every 15 seconds.
- **TTL Sliding Window (45s):** The backend updates the collaborator's last seen epoch. If no heartbeat arrives within 45 seconds, the user is marked offline and stage locks are released.
- **GPS & Geolocation Metadata:** Browser GPS location (latitude/longitude and city/country) is captured once on authenticated sessions and appended via ``X-Browser-Location`` headers for audit forensics.
- **Collision-Avoidance Locking:** When a member actively modifies a stage form, peer clients receive real-time edit alerts and locking indicators, eliminating race conditions.

4. Member Lifecycle & Forensic Audit Trail Engine

Enterprise compliance demands complete historical traceability of who worked on a project, when they joined, when they departed, and when leadership roles were handed over.

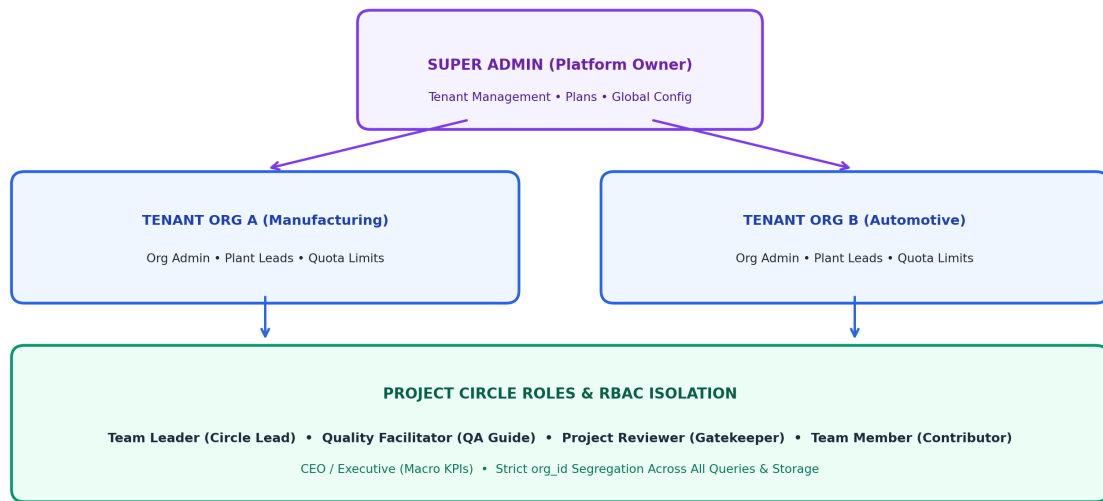


Audit Trail & Lifecycle Capabilities:

- **Inception Baseline Roster:** Automatically records all initial members, facilitators, and reviewers from project inception.
- **Mid-Project Departures & Tenure Tracking:** Detects when members leave the project mid-way. Computes their exact active working duration (e.g. *Joined: Aug 10, 2026 10:00 -> Left: Aug 24, 2026 16:35 [14 Days, 6 Hours active tenure]*) and who removed them.
- **Stakeholder Handover Detection:** Tracks Facilitator replacements (e.g. *in_project_facilitator -> Anita Das*) and Reviewer transitions with separate historical tenure cards and active handover badges.
- **Zero Raw JSON Presentation:** Built-in UI formatter renders delta records as structured Before/After metric comparison tables without raw JSON braces.

5. Multi-Tenant SaaS Governance, Quotas & RBAC Matrix

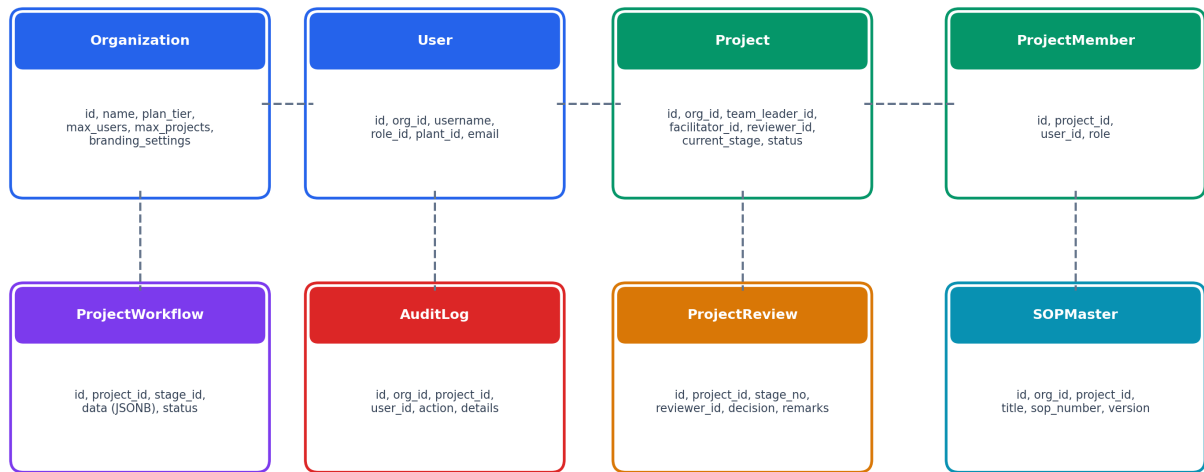
QCCMS provides Total Tenant Isolation, Dynamic Subscription Quota Enforcement, customizable document identity branding, and granular 7-tier role-based access control.



Role Name	Scope & Focus	Key Permissions & Capabilities
Super Admin	Platform Governance	Tenant provisioning, plan limit matrix, system telemetry, document branding, global announcements.
Org Admin	Tenant Operations	User directory, bulk CSV export/import, plant/dept configuration, organization audit stream.
CEO / Executive	Macro Analytics	Read-only access to executive KPI dashboard, financial savings velocity, company-wide circle progress.
Quality Facilitator	QA Guidance	Circle methodology guidance, facilitator stage notes, review recommendation, progress tracking.
Project Reviewer	Gatekeeper Signoff	Independent stage gate evaluation, Approved / Revision / Rejected decisions, audit remarks.
Team Leader	Circle Execution	Project initialization, team formation, 8-stage data entry, countermeasure assignment, final submit.
Team Member	Circle Contributor	Task execution, RCA brainstorming inputs, data collection entry, live presence collaboration.

6. Data Persistence & Relational Entity Topology

The database architecture consists of 85 relational entities supported through SQLAlchemy ORM, organized to enforce multi-tenant organization scoping, fast time-series queries, and robust JSONB workflow payloads.



Core Relational Tables & Schema Purpose:

- **organizations & subscription_plans:** Tenant accounts, domain configuration, plan tier (Starter, Pro, Enterprise, Custom), quota ceilings (max users, max projects, storage MB), and custom branding metadata.
- **users, roles & permissions:** User accounts with bcrypt password hashes, org scoping (`org\_id`), role references (`role\_id`), plant assignments (`plant\_id`), and active presence timestamps.
- **projects & project_members:** Central Quality Circle records with current stage pointers (1-8), status (Active, Review, Closed), and circle stakeholder links (Team Leader, Facilitator, Reviewer).
- **project_workflows:** Stores structured stage data in flexible JSONB format (`data`), capturing 5W2H, Fishbone branches, 5-Why chains, Pareto series, and Countermeasure matrices.
- **project_stage_trackers & project_reviews:** Stage lifecycle trackers (start date, completion date) and official reviewer decisions (Approved, Revision, Remarks, Decided At).
- **audit_logs & facilitator_notes:** Immutable audit telemetry storing user action, IP, target table, timestamp, and JSON delta payloads.

7. Reporting, Document & Automated PDF Publishing Pipeline

QCMS includes an enterprise document generation engine (JasperReports, PyPDF2, and OpenPyXL), producing publication-grade deliverables directly from database state.

Key Document Deliverables:

- **Multi-Page QC Story Closure Report:** Automated creation of comprehensive, publication-ready PDF books detailing all 8 DMAIC stages, high-resolution Fishbone charts, before/after Pareto graphs, photographic evidence tables, team attendance rosters, and digital review signatures.
- **ISO 9001 Compliance Certificates:** Instant vector-rendered certification documents with customized company branding, project UID, team names, and official QR code verification stamps upon Stage 8 closure.
- **Bulk CSV / Excel Data Exporters:** High-speed asynchronous export of user directories, project analytics rosters, and audit trails with custom filter parameters.
- **Tax Invoices & Billing Receipts:** Automatic generation of branded PDF invoices with GST calculations, HSN codes, and tenant billing addresses.

8. Dynamic Multilingual i18n Translation Engine

To empower diverse factory floor workforces across India, QCMS features an intelligent client-side translation engine supporting 6 languages: **English, Hindi, Kannada, Telugu, Tamil, and Malayalam.**

i18n Architecture Highlights:

- **Live DOM MutationObserver:** Continuously monitors dynamic DOM additions (AJAX loaded tables, modal popups, KPI badges) and automatically translates text nodes against language dictionaries in real time.
- **Zero-Flicker Language Switching:** User language preferences are saved in `localStorage` and synchronized across all open browser tabs instantly.
- **Extensible JSON Dictionaries:** Translations are organized in modular dictionary files under `frontend/assets/translations/` for seamless terminology updates.

9. REST API Catalog & Routing Topology

QCMS Backend REST interface is partitioned into Clean Blueprint Routers under `app/presentation/routes/`: IFQM Quality Circle Platform

Router Blueprint	Base Path	Primary Endpoints & Operations
auth_routes	<code>`/api/auth`</code>	<code>`POST /login`, `GET /me`, `POST /heartbeat`, `POST /refresh`, `POST /reset-password`</code>
project_routes	<code>`/api/projects`</code>	<code>`GET /`, `POST /`, `GET /`, `PUT /`, `POST //stage1/save`...`stage8/save`</code>
analytics_routes	<code>`/api/analytics`</code>	<code>`GET /dashboard`, `GET /project-roster`, `GET /kpi-summary`, `GET /savings-velocity`</code>
admin_routes	<code>`/api/admin`</code>	<code>`GET /users`, `POST /users/bulk-import`, `POST /users/export`, `GET /plants`, `GET /audit/logs`</code>
super_admin_routes	<code>`/api/super-admin`</code>	<code>`GET /orgs`, `POST /orgs`, `GET /plans`, `PUT /settings`, `POST /announcements`</code>
sop_routes	<code>`/api/sop`</code>	<code>`GET /masters`, `POST /create`, `GET /`, `PUT /`, `POST //version`</code>
reports_routes	<code>`/api/reports`</code>	<code>`GET /export/pdf/`, `GET /export/excel`, `GET /export/csv`, `GET /certificate/`</code>
workflow_routes	<code>`/api/workflow`</code>	<code>`POST //stage//presence-heartbeat`, `POST //review/submit`</code>

10. Deployment, Docker & Operational Runbook

QCMS is containerized with Docker Compose for seamless cloud and on-premise industrial deployment.

Operational Commands Reference:

- **Docker Multi-Container Launch:** ``docker-compose up --build -d`` (Launches Web UI on :80 and Flask API on :5000).
- **Frontend Asset Minification & Bundling:** ``node build.js`` (Generates cache-busted ``.min.js`` and ``.min.css`` bundles with content hashes).
- **Database Migrations:** ``flask db upgrade`` (Applies Alembic schema revisions safely).
- **Cold Start Verification & Healthcheck:** ``curl http://127.0.0.1:5000/api/auth/maintenance-status`` returns HTTP 200 OK.

Knowledge Transfer Sign-off:

This document serves as the official Knowledge Transfer (KT) and System Architecture specification for QCMS Enterprise OS. Maintained by the core engineering and architecture team for IFQM.